AI in Context: AI 2.0: The Rise of Learning

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Fall 2025 - Lecture 4 of 4

Reminder of Lectures 1, 2 & 3

Lecture 1: Intelligence (esp Human)

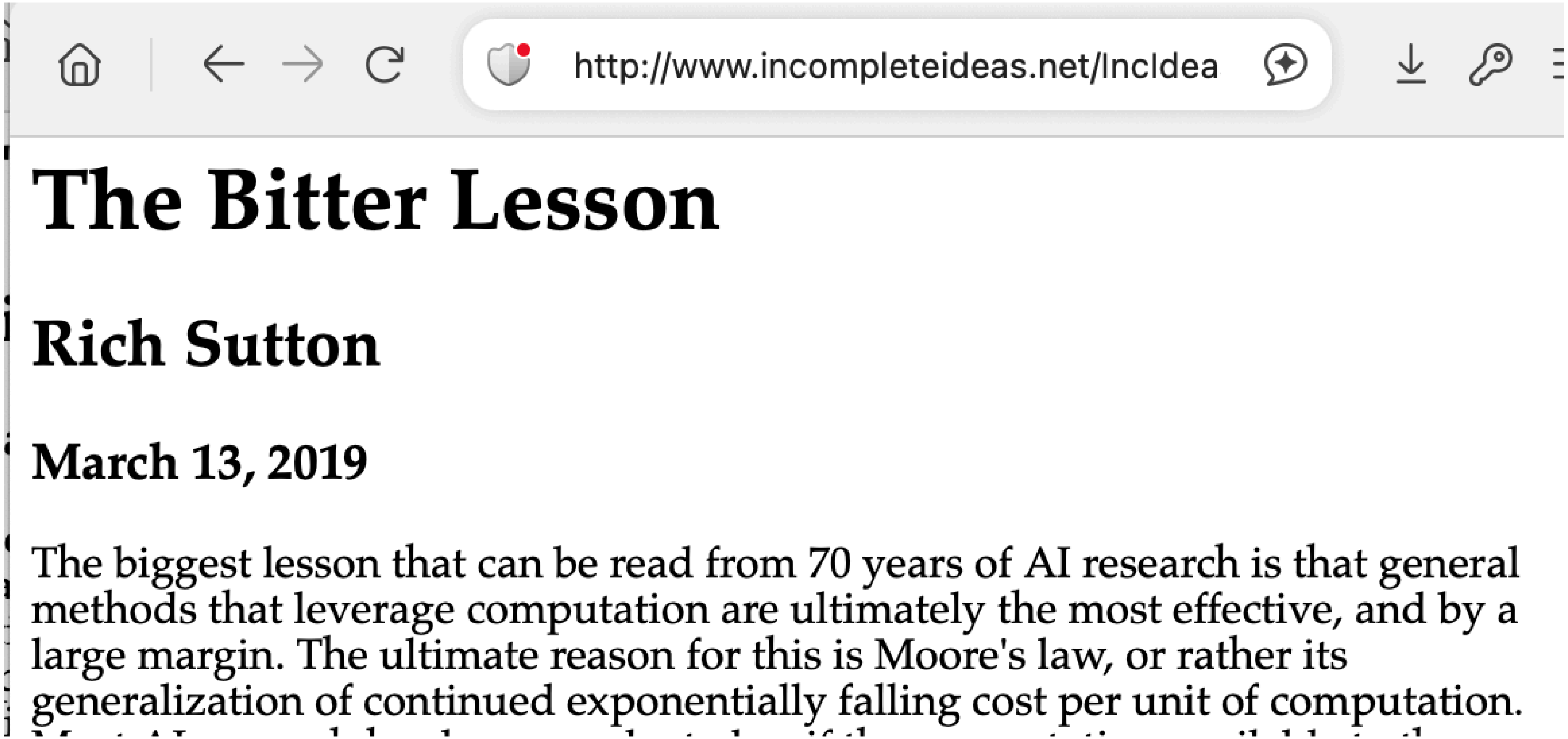
Lecture 2: Dawn of AI (math, logic)

Lecture 3: Birth and death of AI 1.0 (math/interpretability)

Lecture 4: AI 2.0 (data/prediction)

- Part 1: ML 1959-2012: subaltern
- Part 2: ML 2013-2022: "The Bitter Lesson"
- Part 3: ML 2023-present: the inescapable rebrand as "AI"

"The Bitter Lesson", 2019



The Bitter Lesson

Rich Sutton

March 13, 2019

The biggest lesson that can be read from 70 years of AI research is that general methods that leverage computation are ultimately the most effective, and by a large margin. The ultimate reason for this is Moore's law, or rather its generalization of continued exponentially falling cost per unit of computation.

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- Part 1.9: infrastructure: data, compute, scale
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"ML" (the term): Origin 1959 (and 1953)

Some Studies in Machine Learning Using the Game of Checkers

Arthur L. Samuel

Abstract: Two machine-learning procedures have been investigated in some detail using the game of checkers. Enough work has been done to verify the fact that a computer can be programmed so that it will learn to play a better game of checkers than can be played by the person who wrote the program. Furthermore, it can learn to do this in a remarkably short period of time (8 or 10 hours of machine-playing time) when given only the rules of the game, a sense of direction, and a redundant and incomplete list of parameters which are thought to have something to do with the game, but whose correct signs and relative weights are unknown and unspecified. The principles of machine learning verified by these experiments are, of course, applicable to many other situations.

1953: "It may now be worthwhile to talk of a ***machine learning*** process as one in which the machine adapts its code of instructions to conform to its experience." Floyd Steele, ACM conference panel, LA

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Samuel, Arthur L. "Some studies in machine learning using the game of checkers." IBM Journal of research and development 3, no. 3 (1959): 210-229.

NB: He had demo'ed on TV in 1956-02-24 using IBM 701

Part 1: The Dawn of Learning 1950s

- 1956 Samuel @ IBM: Checkers: "He completed the first checker program on the 701, and when it was about to be demonstrated, Thomas J. Watson Sr., the founder and President of IBM, remarked that the demonstration would raise the price of IBM stock 15 points. It did." (-- JCM)
- 1958 recall Rosenblatt @ Cornell Aeronautical Laboratory / ONR : Perceptron
  - see also 1958 NYT article ("NEW NAVY DEVICE LEARNS BY DOING")
- both published in 1959

Descendent of ML 1959: Pattern Recognition 1968

- Nagy, student of Frank Rosenblatt (perceptron 1959)

> *"Pattern recognition is hardly a cohesive discipline in its own right...."*

> *"The performance of the machine is then evaluated on a test set comprising samples not included in the training set."*

Like AI and early ML, the aspiration was (at least initially) implemented via whatever worked: ANNs, statistics, etc

"Pattern Recognition" methods overtook "ML"

> *It was called pattern recognition then; it's called machine learning now.*

- Jerry Friedman, interviewed in 2015, describing work done in 1972

"Pattern Recognition" book (1973) preview of 2010s ML

Duda Hart 1973

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Duda Hart 1973

Pattern Recognition: problems & funders & .mil

From Nils Nilsson's "QAI"; see also "Siri"

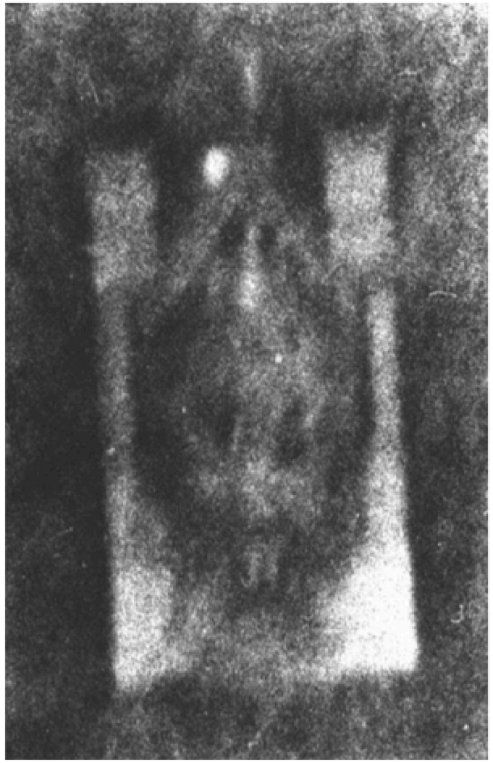


Figure 4.13: A typical tank image. (Photograph courtesy of Thomas Harley.)

ML as "driftword": contrast 1983 ML

"*Historically*, researchers have taken two approaches to machine learning. Numerical methods such as discriminant analysis have proven quite useful in perceptual domains, and have become associated with the paradigm known as *Pattern Recognition*. In contrast, *Artificial Intelligence* researchers have concentrated on *symbolic* learning methods -- Pat Langley

- "parent fields": artificial intelligence and cognitive science
- missing: stats

fast forward 1997 Mitchell on ML

> *"Machine Learning is the study of computer algorithms that improve automatically through experience."*

From http://www.cs.cmu.edu/~tom/mlbook.html

(recall Mitchell role in Simon's "Why should machines Learn?")

Fast forward to 2009-2014 ML as Statistics

MIJ, summer school talk 2009: "[Machine learning is just statistics]"

MIJ, AMA 2014:

> *Throughout the eighties and nineties, it was striking how many times people working within the "ML community" realized that their ideas had had a lengthy pre-history in statistics.*

MIJ 2014 continued

> *Are the SVM and boosting [two prominent techniques 1990s/2000s] machine learning while logistic regression is statistics, even though they're solving essentially the same optimization problems up to slightly different shapes in a loss function? Why does anyone think that these are meaningful distinctions?*

ML, 2011

Mach Learn (2011) 82: 275–279
DOI 10.1007/s10994-011-5242-y

EDITORIAL

The changing science of machine learning

Pat Langley

Changing nature of ML, 2000s

Langley 2011 on ML and what was lost (1/n)

> *early [ML] emphasis on 'symbolic' representations of learned knowledge, such as production rules, decision trees, and logical formulae...discouraged papers on neural networks and other 'nonsymbolic' approaches....the machine learning movement was an outgrowth of symbolic AI and cognitive science, with most of these researchers being concerned with automatically constructing expert systems or modeling human acquisition of knowledge structures.*

Langley 2011 on ML (2/n)

> *by the late 1980s the journal had begun to consider and publish papers on other approaches to learning, some of them incorporating ideas from the pattern-recognition community, which had been exploring the topic for over two decades.*

Langley 2011 on ML (3/n)

> *shift away from the field's original concern with learning in the context of intelligent systems*

- declining centrality of "intelligence"
- rise of statistical methods via "pattern recognition" community

Langley 2011 on ML (4/n)

> *methods borrowed from the pattern-recognition community lent themselves to these simpler tasks [classification, regression]*

- declining centrality of "intelligence"
- rise of statistical methods via "pattern recognition" community

Langley 2011 on ML (5/n)

> *In 1986, when we launched the journal, machine learning was still viewed as a branch of artificial intelligence. By 2000, many researchers committed to machine learning treated it as a separate field with few links to its parent discipline.*

- declining centrality of "intelligence"
- rise of statistical methods via "pattern recognition" community

Langley 2011 on ML (6/n)

> *There are now active PhD-level researchers who have never taken a course in artificial intelligence and who see no reason why they should, as their interests lie in completely different areas.*

- declining centrality of "intelligence"
- rise of statistical methods via "pattern recognition" community

Langley 2011 on ML (7/n) "performance" vs "knowledge"

> *During the 1990s ... [there was a] growing use of statistical and pattern-recognition approaches, which improved performance but which did not produce knowledge in any generally recognized form.*

Langley 2011 on ML (8/n): mathematatization, yet...

Langley 2011 on ML (9/n) ...performance papers

> *most studies involving 'bake offs', that is, mindless comparisons among the performance of algorithms that reveal little about the sources of power or the effects of domain characteristics*

- performance-centrism
- cf. Leaderboard
- and yet... this has had tremendous impact!

Langley 2011 on ML (10/n)

> *Machine learning was originally concerned with developing intelligent systems that exhibited rich behavior on complex tasks, while many modern researchers seem content to tackle problems that do not require either intelligence or systems. Machine learning focused initially on using and acquiring knowledge cast as rich relational structures, while many researchers now appear to care only about statistics.*

Who put "statistics" in my "intelligent systems"?

ML, 2015

REVIEW

Machine learning: Trends, perspectives, and prospects

M. I. Jordan[1*] and T. M. Mitchell[2*]

Machine learning addresses the question of how to build computers that improve automatically through experience. It is one of today's most rapidly growing technical fields, lying at the intersection of computer science and statistics, and at the core of artificial intelligence and data science. Recent progress in machine learning has been driven both by the development of new learning algorithms and theory and by the ongoing explosion in the availability of online data and low-cost computation. The adoption of data-intensive machine-learning methods can be found throughout science, technology and commerce, leading to more evidence-based decision-making across many walks of life, including health care, manufacturing, education, financial modeling, policing, and marketing.

achine learning is a discipline focused

ance when executing some task, through some

2015

Happy synthesis: Jordan-Mitchell 2015

"Machine learning is a discipline focused on two interrelated questions:

1. How can one construct computer systems that automatically improve through experience?
2. What are the fundamental statistical-computational-information-theoretic laws that govern all learning systems, including computers, humans, and organizations?"

J-M 2015:

> *"machine-learning algorithms can be viewed as searching through a large space of candidate programs, guided by training experience, to find a program that optimizes the performance metric."*

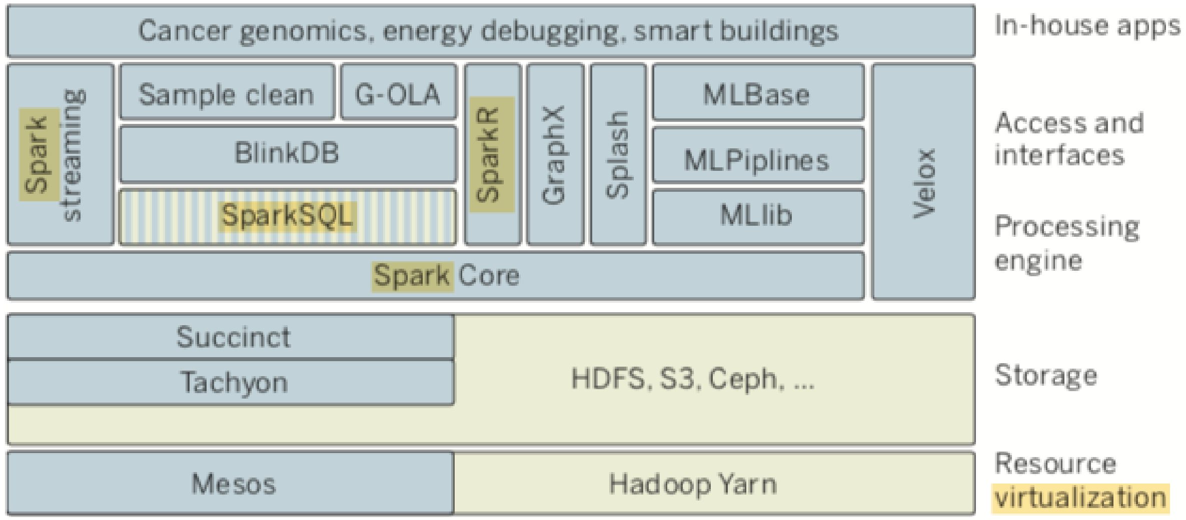
1. "large space of candidate programs" (e.g. different decision trees for classifying)
2. "training experience" (e.g. human classified data)
3. "metric" = what's most important to you (e.g. accuracy)

J-M 2015:

> *Within artificial intelligence machine learning has emerged as the method of choice...it can be far easier to train a system...than to program it*

pre-AGI ML: Key ideas in Machine Learning 2010s

- "three kinds of ML"
- prediction culture
- math $\in$ tech $\in$ socio-technical systems (including ADS)
- impact



Scale, politics, and hints of ethics

- "Granular, personalized...data"
- "minimize privacy effects"
- "differential privacy"
- "social, legal, political framework surrounding the deployment of a system...cooperative or adversarial"

J-M on ethics

> *As with any powerful technology, machine learning raises questions about which of its potential uses society should encourage and discourage. The push in recent years to collect new kinds of personal data, motivated by its economic value, leads to obvious privacy issues, as mentioned above.*

J-M on ethics (continued)

> *The increasing value of data also raises a second ethical issue: Who will have access to, and ownership of, online data, and who will reap its benefits? Currently, much data are collected by corporations for specific uses leading to improved profits, with little or no motive for data sharing. However, the potential benefits that society could realize, even from existing online data, would be considerable if those data were to be made available for public good.*

2010s Key ideas in Machine Learning

ML prediction culture, recall

> *"Overwhelmingly, machine learning systems are oriented towards one specific task: to make accurate predictions."*

-- SD2019

2010s Key ideas in Machine Learning

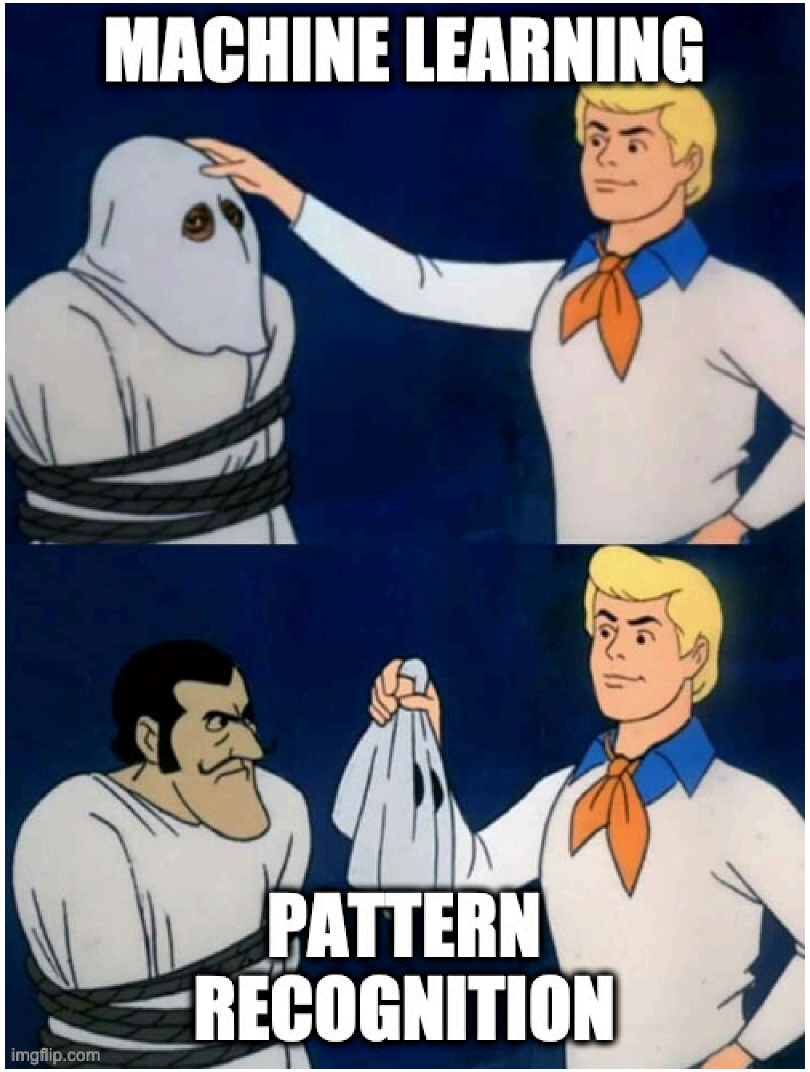
J-M: 3 paradigms

- Supervised (main topic)
- Unsupervised
- Reinforcement

Roughly: predict, describe, prescribe (colab)

History in Memes

History in Memes



History in Memes

History in Memes

Caveat: deep learning is not 1940s stats, which brings us to....

2. "AI 2.0" the re-branding

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Increasing research, PR and popular attention e.g.,

- 2009 – Launch of ImageNet ("common task framework"/prediction central)
- 2011 – Creation of AlexNet
  - error rate was 15.3%; second closest was 26.2%
- 2012 - AlexNet wins imagenet
- 2012 – The Cat Experiment
- 2013 - word2vec
- 2017 - transformers

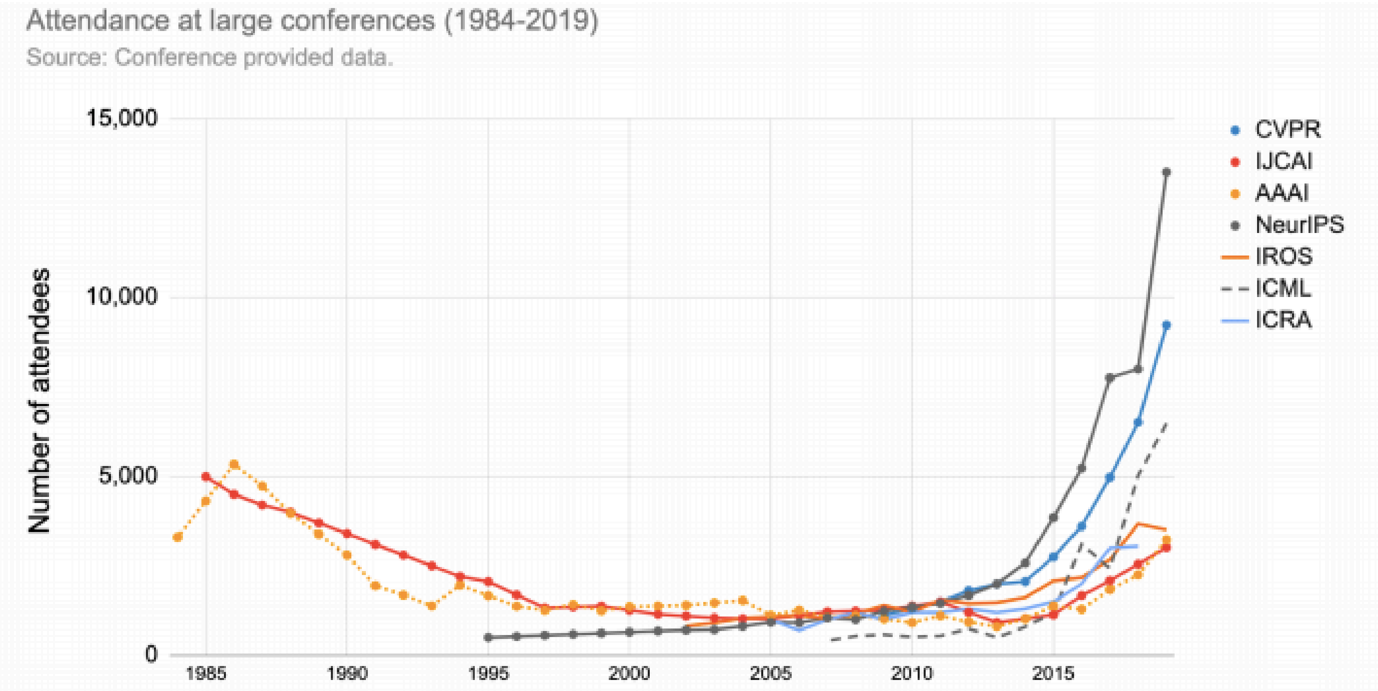

Fig. 2.1a

Note: IJCAI occurred every other year till 2014. The missing year between 1984 and 2014 are interpolated as the mean between the two known conference attendance dates to provide a comparative view across conferences.

From Krauss 2016 NYT:

- 2016
- Why great awakening?
- ML much bigger than MT, but it's a good story
    - e.g., common task framework
    - human-interpretable
    - actual "AI", i.e., recall the "N" of NLP

From Krauss 2016 NYT:

> *Google's decision to reorganize itself around A.I. was the first major manifestation of what has become an industry wide machine-learning delirium. Over the past four years, six companies in particular --- Google, Facebook, Apple, Amazon, Microsoft and the Chinese firm Baidu --- have touched off an arms race for A.I. talent, particularly within universities.*

From Krauss 2016 NYT (continued)

> *Corporate promises of resources and freedom have thinned out top academic departments. It has become widely known in Silicon Valley that Mark Zuckerberg, chief executive of Facebook, personally oversees, with phone calls and video-chat blandishments, his company's overtures to the most desirable graduate students. Starting salaries of seven figures are not unheard-of. Attendance at the field's most important academic conference has nearly quadrupled.*

From Krauss 2016 NYT on alchemy

> *Some of the stuff was not done in full consciousness. They didn't know themselves why they worked.*

cf., Ali Rahimi and Ben Recht, 2017

From Krauss 2016 NYT: Big Data (continued)

> *Neither could Hinton, despite the generosity of the Canadian government. "There just wasn't enough computer power or enough data. People on our side kept saying, 'Yeah, but if I had a really big [computer], it would work.' "It wasn't a very persuasive argument."*

3.0 What was gained? What was lost

- prediction vs modeling
- prediction vs interpretability
- but what *is* interpretability?
    - causality
    - fairness
    - transparency

Prediction vs modeling: Breiman's two cultures

> *"In the mid-1980s two powerful **new** algorithms for fitting data became available: neural nets and decision trees. A new research community using these tools sprang up. Their goal was predictive accuracy. The community consisted of young computer scientists, physicists and engineers plus a few aging statisticians.*

A view from 2022

1. overview of impressive developments since 2015
2. NLP/NLG ; intelligence revisited
3. physical limits: time, money, energy
4. who is doing AI? .edu vs .com (and .us vs .cn)
5. "kinetic"+"autonomy"